

NeetCode 150 - Essential LeetCode Problems for FAANG Interviews

NeetCode 150 is a curated list of 150 essential LeetCode problems compiled by a Google engineer. This collection represents the most important algorithmic patterns and concepts frequently tested in FAANG and top tech company interviews. Each problem has been selected to maximize your interview preparation efficiency.

🎯 Perfect for FAANG Interview Preparation | 🔥 150 Handpicked Problems | ✅
Pattern-Based Learning



Why NeetCode 150?

- ✨ **Curated by Industry Expert:** Selected by a Google engineer based on real interview experience
- 🎯 **Maximum Coverage:** Covers all major algorithmic patterns and data structures
- ⚡ **Efficient Preparation:** Focus on quality over quantity - 150 problems vs. 2000+ on LeetCode
- 📈 **Proven Success:** Used by thousands to land offers at top tech companies
- 🎥 **Video Solutions:** Comprehensive video explanations available for every problem

Problem Categories and Distribution

Category	Problems	Difficulty Distribution	Key Patterns
Arrays & Hashing	9	2 Easy, 6 Medium, 1 Hard	Two Pointers, Hash Maps, Prefix Sums
Two Pointers	5	1 Easy, 4 Medium	Left-Right Pointers, Fast-Slow Pointers
Sliding Window	6	2 Easy, 3 Medium, 1 Hard	Fixed/Variable Window, String Problems
Stack	7	2 Easy, 4 Medium, 1 Hard	Monotonic Stack, Expression Parsing
Binary Search	7	2 Easy, 5 Medium	Search Space Reduction, Peak Finding
Linked List	11	2 Easy, 8 Medium, 1 Hard	Two Pointers, Reversal, Cycle Detection
Trees	15	3 Easy, 10 Medium, 2 Hard	DFS, BFS, Tree Construction
Tries	3	2 Medium, 1 Hard	Prefix Trees, Word Search
Heap/Priority Queue	7	1 Easy, 5 Medium, 1 Hard	K-Elements, Merge Operations
Backtracking	9	6 Medium, 3 Hard	Permutations, Combinations, Subsets
Graphs	13	3 Medium, 10 Hard	DFS, BFS, Union Find, Topological Sort
Advanced Graphs	6	6 Hard	Shortest Path, MST, Advanced Algorithms
1-D Dynamic Programming	12	1 Easy, 8 Medium, 3 Hard	State Transitions, Optimization
2-D Dynamic Programming	11	8 Medium, 3 Hard	Grid DP, String DP
Greedy	8	2 Easy, 4 Medium, 2 Hard	Local Optimal Choices

Category	Problems	Difficulty Distribution	Key Patterns
Intervals	5	1 Easy, 3 Medium, 1 Hard	Merging, Scheduling
Math & Geometry	8	3 Easy, 4 Medium, 1 Hard	Number Theory, Computational Geometry
Bit Manipulation	7	3 Easy, 4 Medium	Bitwise Operations, XOR Tricks

Complete Problem List

Arrays & Hashing (9 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
1	Contains Duplicate	Easy	Hash Set	All FAANG
2	Valid Anagram	Easy	Character Count	All FAANG
3	Two Sum	Easy	Hash Map	All FAANG
4	Group Anagrams	Medium	Hash Map, Sorting	Google, Amazon
5	Top K Frequent Elements	Medium	Heap, Hash Map	All FAANG
6	Product of Array Except Self	Medium	Prefix/Suffix Product	Meta, Apple
7	Valid Sudoku	Medium	Hash Set, Matrix	Apple, Amazon
8	Encode and Decode Strings	Medium	String Manipulation	Google, Meta
9	Longest Consecutive Sequence	Hard	Union Find, Hash Set	Google, Meta

Two Pointers (5 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
10	Valid Palindrome	Easy	Two Pointers	Meta, Apple
11	Two Sum II - Input Array Is Sorted	Medium	Two Pointers	Amazon, Google
12	3Sum	Medium	Two Pointers, Sort	All FAANG
13	Container With Most Water	Medium	Two Pointers	All FAANG
14	Trapping Rain Water	Hard	Two Pointers	Amazon, Google

Sliding Window (6 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
15	Best Time to Buy and Sell Stock	Easy	Single Pass	All FAANG
16	Longest Substring Without Repeating Characters	Medium	Variable Window	All FAANG
17	Longest Repeating Character Replacement	Medium	Variable Window	Google, Meta
18	Permutation in String	Medium	Fixed Window	Meta, Amazon
19	Minimum Window Substring	Hard	Variable Window	Meta, Amazon
20	Sliding Window Maximum	Hard	Deque, Fixed Window	Amazon, Google

Stack (7 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
21	Valid Parentheses	Easy	Stack Matching	All FAANG

No.	Problem	Difficulty	Pattern	Company Tags
22	Min Stack	Medium	Stack Design	Amazon, Apple
23	Evaluate Reverse Polish Notation	Medium	Stack Evaluation	Amazon, Meta
24	Generate Parentheses	Medium	Backtracking, Stack	Google, Meta
25	Daily Temperatures	Medium	Monotonic Stack	Amazon, Google
26	Car Fleet	Medium	Stack, Sorting	Tesla, Amazon
27	Largest Rectangle in Histogram	Hard	Monotonic Stack	Google, Amazon

Binary Search (7 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
28	Binary Search	Easy	Basic Binary Search	All FAANG
29	Search a 2D Matrix	Medium	Matrix Binary Search	Amazon, Apple
30	Koko Eating Bananas	Medium	Binary Search on Answer	Google, Meta
31	Find Minimum in Rotated Sorted Array	Medium	Rotated Array	All FAANG
32	Search in Rotated Sorted Array	Medium	Rotated Array	All FAANG
33	Time Based Key-Value Store	Medium	Binary Search, Design	Google, Amazon
34	Median of Two Sorted Arrays	Hard	Binary Search, Arrays	Google, Amazon

Linked List (11 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
35	Reverse Linked List	Easy	Iterative/Recursive	All FAANG
36	Merge Two Sorted Lists	Easy	Two Pointers	All FAANG
37	Reorder List	Medium	Multiple Pointers	Meta, Amazon
38	Remove Nth Node From End of List	Medium	Two Pointers	Amazon, Google
39	Copy List with Random Pointer	Medium	Hash Map, Cloning	Meta, Amazon
40	Add Two Numbers	Medium	Simulation	Meta, Amazon
41	Linked List Cycle	Easy	Floyd's Algorithm	All FAANG
42	Find the Duplicate Number	Medium	Floyd's Algorithm	All FAANG
43	LRU Cache	Medium	Design, Hash Map	All FAANG
44	Merge k Sorted Lists	Hard	Divide & Conquer	All FAANG
45	Reverse Nodes in k-Group	Hard	Advanced Reversal	Meta, Google

Trees (15 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
46	Invert Binary Tree	Easy	DFS/BFS	All FAANG
47	Maximum Depth of Binary Tree	Easy	DFS/BFS	All FAANG
48	Diameter of Binary Tree	Easy	DFS	All FAANG
49	Balanced Binary Tree	Easy	DFS	Amazon, Apple
50	Same Tree	Easy	DFS/BFS	All FAANG

No.	Problem	Difficulty	Pattern	Company Tags
51	Subtree of Another Tree	Easy	DFS	Amazon, Meta
52	Lowest Common Ancestor of a Binary Search Tree	Medium	BST Property	All FAANG
53	Binary Tree Level Order Traversal	Medium	BFS	All FAANG
54	Binary Tree Right Side View	Medium	BFS/DFS	Amazon, Meta
55	Count Good Nodes in Binary Tree	Medium	DFS	Amazon, Google
56	Validate Binary Search Tree	Medium	DFS, BST	All FAANG
57	Kth Smallest Element in a BST	Medium	Inorder Traversal	All FAANG
58	Construct Binary Tree from Preorder and Inorder Traversal	Medium	Tree Construction	All FAANG
59	Binary Tree Maximum Path Sum	Hard	DFS, Path Sum	All FAANG
60	Serialize and Deserialize Binary Tree	Hard	Tree Serialization	All FAANG

Tries (3 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
61	Implement Trie (Prefix Tree)	Medium	Trie Design	All FAANG
62	Design Add and Search Words Data Structure	Medium	Trie, Backtracking	Google, Meta
63	Word Search II	Hard	Trie, Backtracking	All FAANG

Heap / Priority Queue (7 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
64	Kth Largest Element in a Stream	Easy	Min Heap	All FAANG
65	Last Stone Weight	Easy	Max Heap	Amazon, Apple
66	K Closest Points to Origin	Medium	Min Heap	All FAANG
67	Kth Largest Element in an Array	Medium	Quickselect, Heap	All FAANG
68	Task Scheduler	Medium	Max Heap, Greedy	All FAANG
69	Design Twitter	Medium	Heap, Design	Amazon, Meta
70	Find Median from Data Stream	Hard	Two Heaps	All FAANG

Backtracking (9 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
71	Subsets	Medium	Backtracking	All FAANG
72	Combination Sum	Medium	Backtracking	All FAANG
73	Permutations	Medium	Backtracking	All FAANG
74	Subsets II	Medium	Backtracking, Duplicates	Amazon, Meta
75	Combination Sum II	Medium	Backtracking, Duplicates	Amazon, Google
76	Word Search	Medium	Backtracking, Matrix	All FAANG
77	Palindrome Partitioning	Medium	Backtracking, String	Amazon, Google

No.	Problem	Difficulty	Pattern	Company Tags
78	Letter Combinations of a Phone Number	Medium	Backtracking	All FAANG
79	N-Queens	Hard	Backtracking	Google, Amazon

Graphs (13 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
80	Number of Islands	Medium	DFS/BFS	All FAANG
81	Clone Graph	Medium	DFS/BFS, Cloning	All FAANG
82	Max Area of Island	Medium	DFS	Amazon, Google
83	Pacific Atlantic Water Flow	Medium	DFS/BFS	Google, Amazon
84	Surrounded Regions	Medium	DFS/BFS	Amazon, Google
85	Rotting Oranges	Medium	BFS	Amazon, Apple
86	Walls and Gates	Medium	BFS	Google, Meta
87	Course Schedule	Medium	Topological Sort	All FAANG
88	Course Schedule II	Medium	Topological Sort	All FAANG
89	Redundant Connection	Medium	Union Find	Google, Amazon
90	Number of Connected Components in an Undirected Graph	Medium	Union Find/DFS	Google, Meta
91	Graph Valid Tree	Medium	Union Find/DFS	Google, Meta

No.	Problem	Difficulty	Pattern	Company Tags
92	Word Ladder	Hard	BFS	All FAANG

Advanced Graphs (6 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
93	Reconstruct Itinerary	Hard	Eulerian Path	Google, Amazon
94	Min Cost to Connect All Points	Medium	MST (Prim's/Kruskal's)	Amazon, Google
95	Network Delay Time	Medium	Dijkstra's Algorithm	Amazon, Google
96	Swim in Rising Water	Hard	Dijkstra's/Binary Search	Google, Amazon
97	Alien Dictionary	Hard	Topological Sort	Google, Meta
98	Cheapest Flights Within K Stops	Medium	Bellman-Ford/Dijkstra's	Google, Amazon

1-D Dynamic Programming (12 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
99	Climbing Stairs	Easy	Basic DP	All FAANG
100	Min Cost Climbing Stairs	Easy	Basic DP	Amazon, Apple
101	House Robber	Medium	Linear DP	All FAANG
102	House Robber II	Medium	Circular Array DP	All FAANG
103	Longest Palindromic Substring	Medium	String DP	All FAANG
104	Palindromic Substrings	Medium	String DP	Amazon, Google

No.	Problem	Difficulty	Pattern	Company Tags
105	Decode Ways	Medium	String DP	All FAANG
106	Coin Change	Medium	DP, Greedy	All FAANG
107	Maximum Product Subarray	Medium	DP	All FAANG
108	Word Break	Medium	String DP	All FAANG
109	Longest Increasing Subsequence	Medium	DP, Binary Search	All FAANG
110	Partition Equal Subset Sum	Medium	Subset Sum DP	Amazon, Google

2-D Dynamic Programming (11 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
111	Unique Paths	Medium	Grid DP	All FAANG
112	Longest Common Subsequence	Medium	String DP	All FAANG
113	Best Time to Buy and Sell Stock with Cooldown	Medium	State Machine DP	Google, Amazon
114	Coin Change 2	Medium	Unbounded Knapsack	Amazon, Google
115	Target Sum	Medium	Subset Sum DP	Google, Meta
116	Interleaving String	Medium	String DP	Google, Amazon
117	Longest Increasing Path in a Matrix	Hard	DFS + Memoization	Google, Amazon
118	Distinct Subsequences	Hard	String DP	Google, Amazon
119	Edit Distance	Hard	String DP	Google, Meta

No.	Problem	Difficulty	Pattern	Company Tags
120	Burst Balloons	Hard	Interval DP	Google, Amazon
121	Regular Expression Matching	Hard	String DP	All FAANG

Greedy (8 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
122	Maximum Subarray	Medium	Kadane's Algorithm	All FAANG
123	Jump Game	Medium	Greedy	All FAANG
124	Jump Game II	Medium	Greedy	All FAANG
125	Gas Station	Medium	Greedy	Amazon, Google
126	Hand of Straights	Medium	Greedy, Hash Map	Google, Amazon
127	Merge Triplets to Form Target Triplet	Medium	Greedy	Google, Amazon
128	Partition Labels	Medium	Greedy	Amazon, Google
129	Valid Parenthesis String	Medium	Greedy	Google, Meta

Intervals (5 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
130	Insert Interval	Medium	Interval Merge	All FAANG
131	Merge Intervals	Medium	Interval Merge	All FAANG
132	Non-overlapping Intervals	Medium	Greedy	Amazon, Google

No.	Problem	Difficulty	Pattern	Company Tags
133	Meeting Rooms	Easy	Sorting	All FAANG
134	Meeting Rooms II	Medium	Heap, Greedy	All FAANG

Math & Geometry (8 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
135	Rotate Image	Medium	Matrix Manipulation	All FAANG
136	Spiral Matrix	Medium	Matrix Traversal	All FAANG
137	Set Matrix Zeroes	Medium	Matrix Modification	Amazon, Apple
138	Happy Number	Easy	Number Theory	Google, Apple
139	Plus One	Easy	Array Manipulation	Google, Apple
140	Pow(x, n)	Medium	Fast Exponentiation	All FAANG
141	Multiply Strings	Medium	String Math	Google, Meta
142	Detect Squares	Medium	Geometry	Google, Amazon

Bit Manipulation (7 Problems)

No.	Problem	Difficulty	Pattern	Company Tags
143	Single Number	Easy	XOR	All FAANG
144	Number of 1 Bits	Easy	Bit Counting	All FAANG
145	Counting Bits	Easy	DP + Bits	All FAANG
146	Reverse Bits	Easy	Bit Reversal	Apple, Amazon
147	Missing Number	Easy	XOR/Math	All FAANG
148	Sum of Two Integers	Medium	Bit Arithmetic	Apple, Amazon

No.	Problem	Difficulty	Pattern	Company Tags
149	Reverse Integer	Medium	Integer Manipulation	All FAANG

Study Strategy

Week 1-2: Foundation (Arrays, Two Pointers, Sliding Window)

- Master basic patterns with arrays and strings
- Focus on understanding time/space complexity
- Practice implementation without looking at solutions

Week 3-4: Data Structures (Stack, Linked Lists, Trees)

- Learn fundamental data structure operations
- Practice traversal algorithms (DFS, BFS)
- Understand when to use each data structure

Week 5-6: Advanced Patterns (Backtracking, Graphs, Heaps)

- Study complex algorithmic patterns
- Practice problem decomposition
- Learn to identify pattern similarities

Week 7-8: Dynamic Programming & Optimization

- Start with 1D DP problems
- Progress to 2D DP and advanced techniques
- Focus on state transition understanding

Week 9-10: Final Topics & Review

- Complete remaining categories
- Review challenging problems
- Take timed practice sessions



Success Tips

1. **Understand Patterns:** Don't just memorize solutions - understand the underlying patterns
2. **Time Yourself:** Practice under time constraints (20-30 minutes per problem)
3. **Code Without IDE:** Practice writing code on paper or whiteboard
4. **Explain Solutions:** Be able to articulate your approach and trade-offs
5. **Review Regularly:** Revisit problems you found challenging
6. **Mock Interviews:** Practice explaining your solution process

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